

Atif Rao

Agricultural and Applied Economist | Climate Econometrics | International Agricultural Trade | Food Security

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Department of Agricultural Economics, Kansas State University, Manhattan, KS, USA

PROFILE AND RESEARCH FIELDS

- **Fulbright Ph.D. candidate in Agricultural Economics** working at the intersection of **climate econometrics and international agricultural trade**. Research quantifies how extreme heat, drought, and flood disasters transmit from field-level production into **market values, export allocations, and bilateral trade flows**, with direct implications for trade competitiveness and food security.
- Methods span **panel fixed-effects estimation, structural gravity with PPML, event-study and heterogeneity designs, and CMIP6-based scenario projection**, applied to large multi-source datasets built from PRISM, CMIP6, EM-DAT, USDA, FAO, and USITC. More than **10 years of research, teaching, and policy experience** across the **United States, China, Pakistan, and Germany**.

Primary fields: Applied econometrics; international agricultural trade; climate and weather economics.

Secondary fields: Agricultural production and productivity; food security and resilience; trade and development policy.

EDUCATION

Kansas State University

Sep 2026

Ph.D. Candidate, Agricultural Economics (Fulbright Scholar)

Manhattan, KS, USA

- Dissertation: *Three Essays on Extreme Weather, Agricultural Markets, and International Trade*. Defense scheduled September 17, 2026.
- Advisor: Dr. Nelson Villoria. Committee: Dr. Peri da Silva, Dr. Nathan Hendricks, Dr. Jisang Yu.

Friedrich Schiller University of Jena

Mar 2015

M.Sc. Economics

Jena, Germany

- Thesis: *An Empirical Analysis of the Impact of FDI on Innovation* – awarded distinction.

PUBLICATIONS AND RESEARCH

Peer-Reviewed Publications

- [P.1] Rao, Atif, Muhammad Ali, and Jason M. Smith (2024). Foreign Direct Investment and Domestic Innovation: Roles of Absorptive Capacity, Quality of Regulations and Property Rights. *PLOS ONE*.

Working Papers

- [W.1] Rao, Atif and Nelson Villoria (2026). Weather Extremes, Production Value, and Export Allocations: Evidence from Corn and Soybeans in the U.S. Midwest. *Revise and resubmit*, Journal of the Agricultural and Applied Economics Association. (**Job market paper**)
- [W.2] Rao, Atif and Nelson Villoria (2026). The Impact of Flood Disasters on Rice Trade Flows in Southeast Asia: Evidence from a Gravity Framework.
- [W.3] Rao, Atif and Nelson Villoria (2026). Floods, Export Policy, and Rice Trade in South Asia: Heterogeneous Responses across Exporters.

RESEARCH EXPERIENCE

Kansas State University

2022–Present

Doctoral Researcher – Climate Economics and Agricultural Trade

Manhattan, KS, USA

- Design and estimate **panel and gravity models** linking extreme weather exposure to crop production value, domestic use, and bilateral trade flows; implement PPML with high-dimensional fixed effects, event-study specifications, and heterogeneity analysis across exporters and policy regimes.
- Construct **large harmonized panels** merging gridded climate data (PRISM, CMIP6), disaster records (EM-DAT), and production and trade data (USDA NASS/ERS, FAO, USITC, UNCTAD); build reproducible pipelines in R, Python, and Stata with automated validation and QA.
- Translate results into **papers, seminar presentations, policy-relevant briefs, tables, and figures** for academic, industry, and policy audiences.

Friedrich Schiller University of Jena

2014–2015

Research Assistant – Economics

Jena, Germany

- Supported empirical research on growth, investment, and innovation through data collection, literature review, quantitative analysis, and preparation of tables, figures, and research notes.

TEACHING EXPERIENCE

Northwest Minzu University
Vice Director and Lecturer of Economics

Mar 2017–Mar 2022
Lanzhou, China

- Designed and taught **Quantitative Methods, Financial Economics, International Finance, Global Economic Issues, and Economics and Law**, including lectures, case-based exercises, assessments, and course materials.
- Co-managed a **joint academic program with Utah State University**, coordinating curriculum standards, scheduling, student advising, and program administration; delivered communication and pedagogy training to roughly 50 faculty and administrators.

Teaching interests: Econometrics and applied quantitative methods; international trade; agricultural and resource economics; environmental and climate economics; principles of microeconomics and macroeconomics.

POLICY AND APPLIED RESEARCH EXPERIENCE

Management Systems International (MSI)
Field Researcher – Program Evaluation and Applied Research

Aug 2016–Jan 2017
Islamabad, Pakistan

- Conducted stakeholder interviews, primary data collection, and instrument design for institutional and program evaluations; coded and synthesized qualitative evidence in **MAXQDA** and contributed to evaluation findings and recommendations.

Sustainable Development Policy Institute (SDPI)
Research Associate – Climate, Agriculture, and Public Investment

Feb 2016–Aug 2016
Islamabad, Pakistan

- Analyzed national and provincial budgets for **climate-related public expenditure**, resource-allocation gaps, and adaptation priorities; contributed to policy briefs on agriculture, rural livelihoods, and sustainable development, and to CPEC analysis covering infrastructure, energy, and environmental governance.

TECHNICAL AND QUANTITATIVE SKILLS

Econometric Methods	Panel data and two-way fixed effects; structural gravity and PPML with high-dimensional fixed effects; event studies; instrumental variables; heterogeneity and robustness analysis; scenario projection.
Programming	R (fixest, data.table, tidyverse, sf), Stata, Python (pandas, xarray, geopandas); Git for version control; $\LaTeX$ and R Markdown for reproducible reporting.
Climate & Spatial Data	PRISM, CMIP6, CRU, ERA5-style gridded products, EM-DAT; raster-to-county aggregation, growing-degree-day and extreme-heat construction, crop-mask weighting.
Trade & Ag Data	USDA NASS/ERS, FAO/FAOSTAT, USITC, UNCTAD, WDI, Statistics Canada balance sheets; bilateral trade concordances and product-code harmonization.
Applied & Qualitative	Power BI, Excel, MAXQDA; stakeholder interviews, qualitative coding, evidence triangulation, program evaluation support, and technical writing for non-technical audiences.

PRESENTATIONS AND PROFESSIONAL ENGAGEMENT

AAEA Annual Meeting – Graduate Case Study Competition, presenting team member	Aug 2025
Risk and Profit Conference, Kansas State University – research presentation	Aug 2025
IMF/World Bank Spring Meetings, Washington, DC – participant	Apr 2026

HONORS AND AWARDS

Fulbright Ph.D. Scholarship, United States Educational Foundation in Pakistan	2022–2026
Third Place, AAEA Graduate Case Study Competition	Aug 2025
Distinction in M.Sc. Thesis, Friedrich Schiller University of Jena	2015

SERVICE AND LEADERSHIP

Vice President, Fulbright Kansas State Group (Aug 2025–Present) – organize academic, professional, and cross-cultural programming for Fulbright scholars.

Vice President, International Buddies Group, Kansas State University (Aug 2024–Aug 2025) – coordinated programs supporting international students and cross-cultural engagement.

LANGUAGES

English: Fluent | Urdu/Hindi: Native | German: Conversational | Arabic: Basic | Mandarin: Basic

REFERENCES

1. Dr. Nelson Villoria, Major Professor, Agriculture Economics Kansas State University, USA | Email: nvilloria@ksu.edu

2. Dr. Peri Da Silva, Professor, Department of Economics, Kansas State University, USA | Email: pdasilva@ksu.edu